TESTS PASSED PROOF AND COVERAGE REPORT

1) Tests passed when running *mvn clean test*

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running HeartRateStrategyTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.047 s -- in HeartRateStrategyTest
[INFO] Running DataReaderTest
File data reading success
File data reading success
File data reading success
Patient data reading failure 1775492803872 low
File data reading success
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.009 s -- in DataReaderTest
[INFO] Running BloodPressureStrategyTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.011 s -- in BloodPressureStrategyTest
[INFO] Running AlertFactoriesTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.005 s -- in AlertFactoriesTest
[INFO] Running PatientRecordsRetrievalTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s -- in PatientRecordsRetrievalTest
[INFO] Running SingletonTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.010 s -- in SingletonTest
[INFO] Running OxygenSaturationStrategyTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.007 s -- in OxygenSaturationStrategyTest
[INFO] Running DataStorageTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in DataStorageTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 20, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

2) Coverage report generated

cardio_generator

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.cardio_generator.generators		0%		0%	24	24	101	101	16	16	5	5
com.cardio_generator		6%		7%	24	27	78	85	11	13	0	1
com.cardio_generator.outputs		1%		0%	17	18	59	60	15	16	4	5
com.data_management		74%		68%	13	38	32	104	4	19	0	4
com.alerts		87%		84%	21	72	31	155	11	33	4	15
com		0%		0%	4	4	5	5	2	2	1	1
Total	1,343 of 2,358	43%	72 of 166	56%	103	183	306	510	59	99	14	31

Report shows that the tests concerning packages *com.alerts* and *com.data_management* containing new classes from parts 3 and 4 were relatively comprehensive. Package com.cardio_generator.generators was not tested at all as the outputs of the methods depend on randomness and thus, it is difficult to assert a deterministic correct outcome. Maybe an output range could be checked to verify the behaviour to some extent. Nonetheless, the class would have to checked multiple times to determine with some degree of certainly that it would not produce an incorrect output as it is non-deterministic. Output strategies (in package com.cardio_generator.outputs) and HealthDataSimulator were similarly untested as they depend on network connections, file I/O, and scheduled threading which are not suitable for unit testing without mocking frameworks.